



Device Management

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Estimation Procedure

Use

If measurement data is incomplete or implausible, it may be necessary to replace or add to this data with estimated values. Data simulations and forecasts are also based on estimations. In SAP Utilities, estimation procedures are used in the following processes:

- Estimate meter reading results
- Validate meter reading results
- Billing simulation
- Budget billing plan creation
- Unbilled revenue reporting

The estimation procedures used are **extrapolation** and **interpolation**. When using the quantity determination procedure [Quantity Determination During Meter Reading](#), both of these procedures use the quantities saved in the system for the respective base period as a basis for estimation. The system uses these quantities to calculate the quantity for the estimation period and then calculates the estimated value for the meter reading.

During extrapolation, a value is determined beyond the base period. During interpolation, a value is estimated between other existing values. The base period is the period that contains all meter readings used as the basis for estimation.

One feature of the quantity determination procedure [Quantity Determination During Meter Reading](#) for the gas division is that the quantity is calculated once an actual meter reading is available following one or more estimations (usually extrapolations). In this case, the gas factors are determined for the most recent actual previous meter reading result. This part of the concept is described in section **Interaction - Estimation and Meter Reading**.

With these differences in mind, you can read the information about the classical estimation procedures for discrete values:

[Classical Estimation Procedure](#)

The aspects specifically relevant for quantity determination during meter reading are described in the following sections.

During estimation, assumptions are made regarding the time-related distribution of the quantity on which the estimation is based. This distribution is described with a weighting function $y(t)$. The examples in the further sections are based on a linear distribution of weightings in which the weighting function increases by one unit per day. In practice, you can also set up and use other weighting methods in the system. For more information about weighting procedures and the necessary Customizing settings, see [Weighting Procedures](#).

Features

The following sections describe extrapolation and interpolation as estimation procedures for discrete data during the use of the quantity determination procedure [Quantity Determination During Meter Reading](#).

[Extrapolation](#)

The concept for calculating the quantity once an actual meter reading result is available following one or more estimations (usually extrapolations) is described in the following section:

[Interaction - Estimation and Actual Meter Reading](#)

Extrapolation

Use

During the estimation procedure of extrapolation, a value is forecast for the entire base period.

The steps described below are carried out automatically by the system and are not visible to the user. The description should explain the algorithm that derives an expected quantity from a quantity stored in the system and forecasts a meter reading result.

Features

When using the quantity determination procedure [Quantity Determination During Meter Reading](#), extrapolation takes place as follows:

1. The saved (and rounded) quantities from the meter reading results in the representative base period are added together as a basis for extrapolation.
2. The system determines the weighting Y of the base period.
3. The system determines the weighting Y_e of the forecast period.
4. The system determines the quantity Q_e for the forecast date by multiplying the added quantity from the base period with the quotient of the [weighting of the forecast period](#) and the [weighting of the base period](#).

$$Q_e = Q * (Y_e / Y) = (Q / Y) * Y_e$$

Formula

The factor (Q / Y) describes the [Expected Weighted Consumption](#).

5. From this forecast quantity Q_e , that has not yet been rounded, the system calculates the estimated meter reading result Q_e taking the relevant factors (such as the gas factors and register factors) into account.

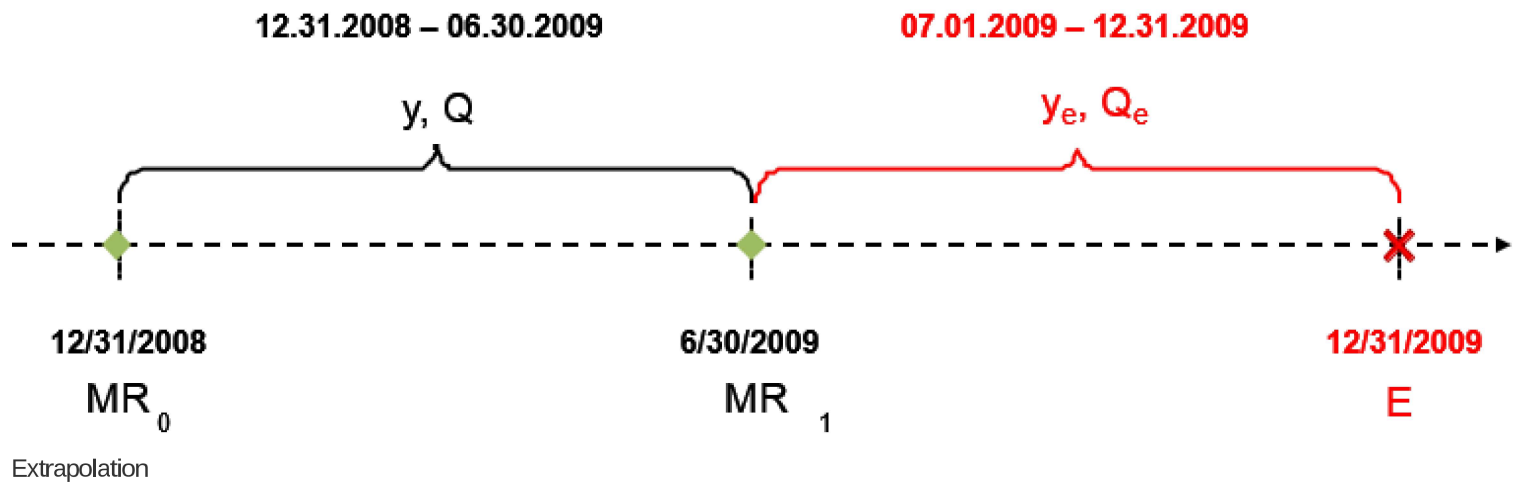
This is then rounded and saved according to the system settings.

6. Taking into account the factors from the forecast period, the system determines the final consumption value from the difference between the estimated meter reading result E and the previous meter reading result $MR1$. The final consumption value is then rounded and saved.

i Note

The system does not wait for actual gas factors for extrapolation, but determines the final quantity immediately. This means that no quantity determination triggers are generated.

- Q_e = Forecasted quantity
- Y = Weighting of base period
- Y_e = Weighting of forecast period
- MR = Meter reading result
- E = Estimated meter reading result



In the example, the base period is between MR_0 and MR_1 ; that is 12.31.2008 to 06.30.2009.

In the example, the forecast period is between MR_1 and E ; that is 07.01.2009 to 12.31.2009.

Constraints

Whereas during the classical quantity determination procedure, you can exclude estimated values from being used as a basis for extrapolation, this is not possible when using the quantity determination procedure [Quantity Determination During Meter Reading](#) as the estimated quantities are also necessary as a basis for extrapolation. If you have selected the [Ignore Estimated Meter Reading Results During Extrapolation](#) checkbox in the control parameters in Customizing for Meter Reading Data Processing, the system ignores this setting as soon as a period exists in which the quantity determination procedure [Quantity Determination During Meter Reading](#) is used for a register.

Customizing for [SAP Utilities](#) > [Device Management](#) > [Meter Reading](#) > [Basic Settings](#) > [Define Control Parameters for Meter Reading Data Processing](#).

Interpolation

Use

During the estimation procedure of interpolation, a value is estimated between two existing values. These existing values can be actual values or estimated values.

The steps described below are carried out automatically by the system and are not visible to the user. The description should explain the algorithm that uses the weighting procedure to distribute a quantity saved in the system and, from that, derives the estimated meter reading result.

When using the quantity determination procedure [Quantity Determination During Meter Reading](#), interpolation between consecutive meter readings takes place as follows:

1. The basis of interpolation between two meter reading results is the quantity saved and rounded for the second meter reading result.
2. The system determines the weighting Y of the base period
3. The system determines the weighting Y_e of the period for which you want the quantity Q_e to be estimated.
4. The system calculates the estimated quantity Q_e as follows:

$$Q_e = Q * (Y_e / Y) = (Q / Y) * Y_e$$

Formula

The factor (Q / Y) describes the [Expected Weighted Consumption](#).

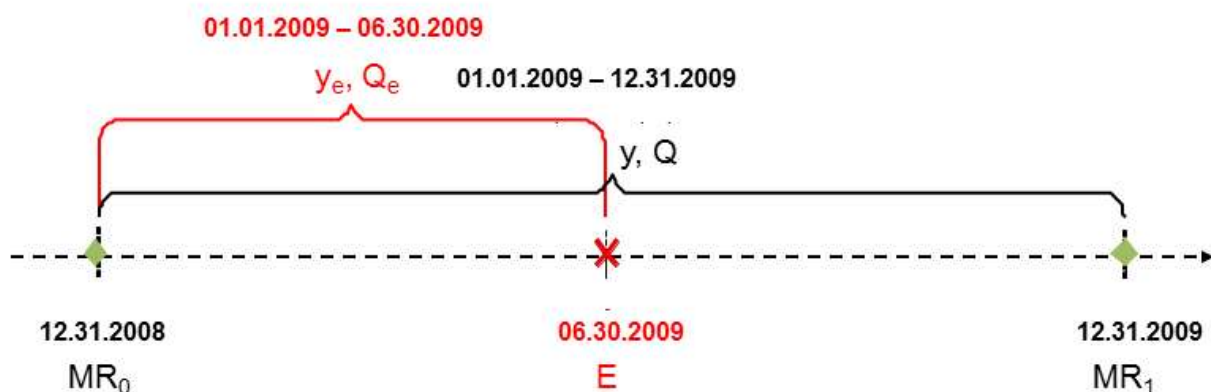
5. From this estimated quantity Q_e , that has not yet been rounded, the system calculates the estimated meter reading result E taking the relevant factors (such as the gas factors and register factors) into account.

This is then rounded and saved according to the system settings.

The following figure shows interpolation using a simple example: The estimated meter reading result E was interpolated based on the estimated quantity Q_e .

The abbreviations have the following meanings:

- MR = Meter Reading Result
- Q = Quantity
- Y = Weighting
- E = Estimated Meter Reading Result
- Q_e = Estimated Quantity
- Y_e = Weighting for Estimated Quantity

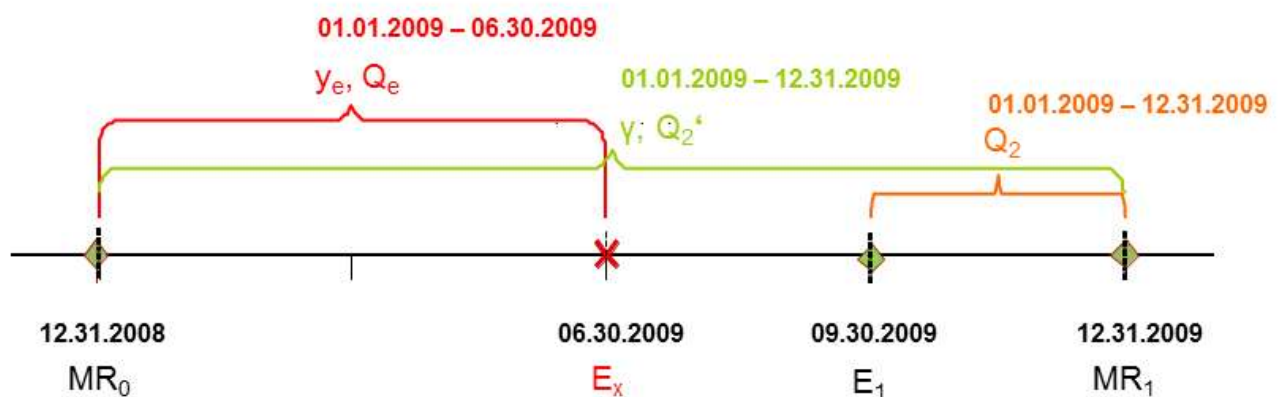


The system prefers actual meter reading results to estimated meter reading results as a basis for estimation. If an estimated meter reading result follows an actual meter reading result, the base period is extended as far as the actual meter reading result.

This is explained in the following figure.

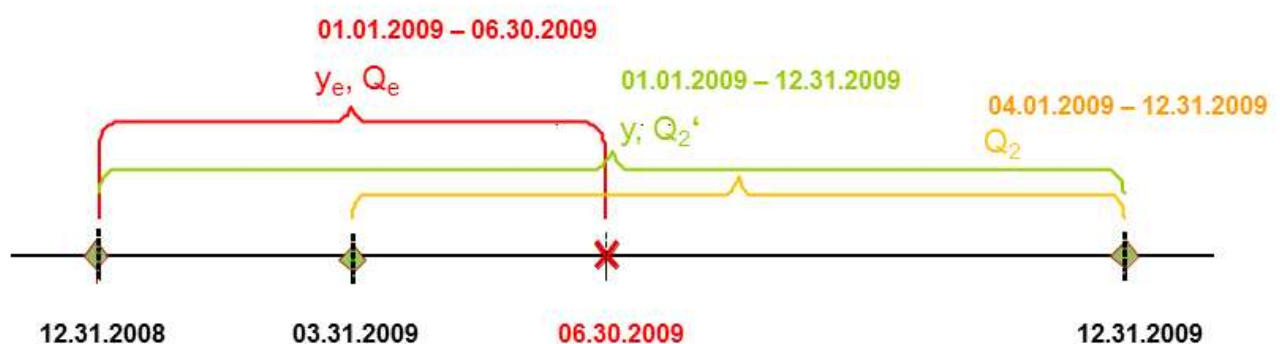
The abbreviations have the following meanings:

- MR = Actual Meter Reading Result
- E = Estimated Meter Reading Result
- Y = Weighting
- Q2 = Quantity for Meter Reading Result MR2
- Q2' = New Quantity Dynamically Calculated for Meter Reading Result MR2 as Part of Interpolation
- Qe = Estimated Quantity
- Ye = Weighting for Estimated Quantity



Example 1

In example 1, there are three meter reading results in the period December 31, 2008 to December 31, 2009: MR₀, E₁, and MR₁. E₁ is an estimated meter reading result. The algorithm ignores this estimated result during interpolation and uses the two actual meter reading results MR₀ and MR₁ instead. As described in the previous section [Interpolation Between Consecutive Meter Reading Results](#), the basis for interpolation is the quantity saved and rounded from the second meter reading result. This quantity Q₂ refers to the period MR₁ to MR₂, which is from October 01, 2009 to December 31, 2009. However, for the interpolation, the system needs a quantity that refers to the period from January 1, 2009 to December 31, 2009. The system calculates this quantity Q₂' dynamically. The next steps for determining the estimated value E_x are identical to steps 2-5 from the previous section [Interpolation Between Consecutive Meter Readings](#).



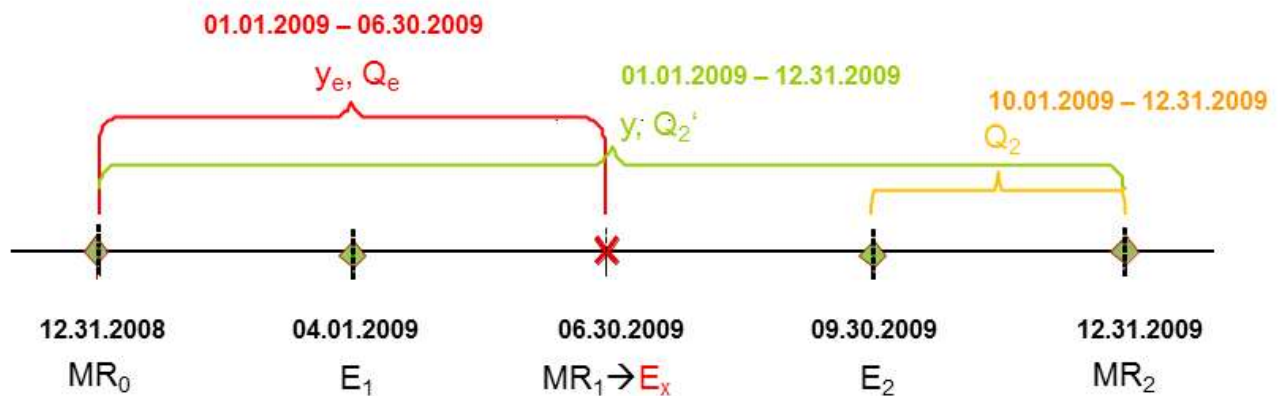
In example 2, there are three meter reading results in the period December 31, 2008 to December 31, 2009: MR0, E1, and MR1. E1 is an estimated meter reading result. The algorithm ignores this estimated result during interpolation and uses the two actual meter reading results MR0 and MR1 instead. The basis of interpolation is the quantity saved and rounded for the second meter reading result. This quantity Q2 refers to the period E1 to MR1, which is from April 1, 2009 to December 31, 2009. However, for the interpolation, the system needs a quantity that refers to the period from January 1, 2009 to December 31, 2009. The system calculates this quantity Q2' dynamically. The next steps for determining the estimated value Ex are identical to steps 2-5 from the section [Interpolation Between Consecutive Meter Readings](#).

⚠ Caution

During quantity determination for the estimated and saved meter reading result Ex, the estimated meter reading result E1 is used. The quantity saved for meter reading result Ex refers to the period up to the previous meter reading result (E1 - Ex).

The base period can be extended in both directions. This is shown in the following example. You want to validate an actual meter reading result MR1, which is between two estimated meter reading results E1 and E2. The estimation procedure of interpolation is also used during validation if you correct meter reading results using transactions EL 27 and EL29.

This is explained in the following figure. The same abbreviations are used as in example 1.



Example 3

In example 3, E1 and E2 are estimated meter reading results. The algorithm ignores these estimated results during interpolation and uses the two actual meter reading results MR0 and MR2 instead. The basis of interpolation is the quantity saved and rounded for the second meter reading result. This quantity Q2 refers to the period E2 to MR2, which is from October 1, 2009 to December 31, 2009. However, for the interpolation, the system needs a quantity that refers to the period from January 1, 2009 to December 31, 2009. The system calculates this quantity Q2' dynamically. The next steps for determining the estimated value E are identical to steps 2-5 from the section [Interpolation Between Consecutive Meter Readings](#). The quantity saved for meter reading result Ex refers to the period up to the previous meter reading result (E1 - Ex).

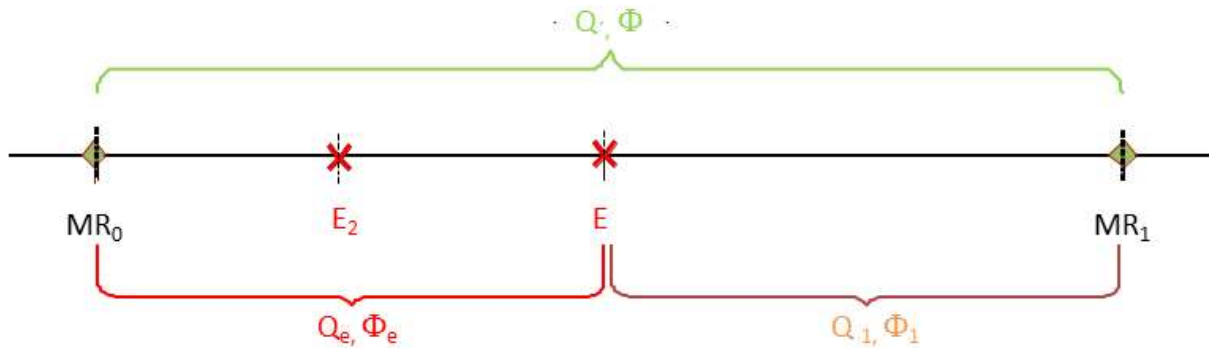
Interpolation Between Estimated Meter Readings

If interpolation or reversal takes place multiple times between estimated meter reading results, it is generally not possible to determine the original total consumption.

This is explained in the following figure. The abbreviations have the following meanings:

- MR = Actual Meter Reading Result
- E = Estimated Meter Reading Result

- Q_e = Estimated Quantity
- Φ = Gas Factor



Example 4

In example 4, interpolation takes place between the estimated meter readings MR_0 and MR_1 . The system uses the (averaged) factor Φ from the period MR_0 to MR_1 in all steps for the conversion between meter reading and consumption. This ensures that the total consumption Q is retained:

$$Q = Q_e + Q_1.$$

If another meter reading is subsequently estimated between MR_0 and E , the system uses the averaged factor Φ_e from the period MR_0 to E for the conversion between the meter readings and consumption values. However, as the factors Φ and Φ_e are generally different, the total consumption Q is not retained.

Interaction - Estimation and Actual Meter Reading

Use

One feature of the quantity determination procedure [Quantity Determination During Meter Reading](#) for the gas division is the way in which the quantity is calculated once an actual meter reading is available following one or more estimations (usually extrapolations).

Features

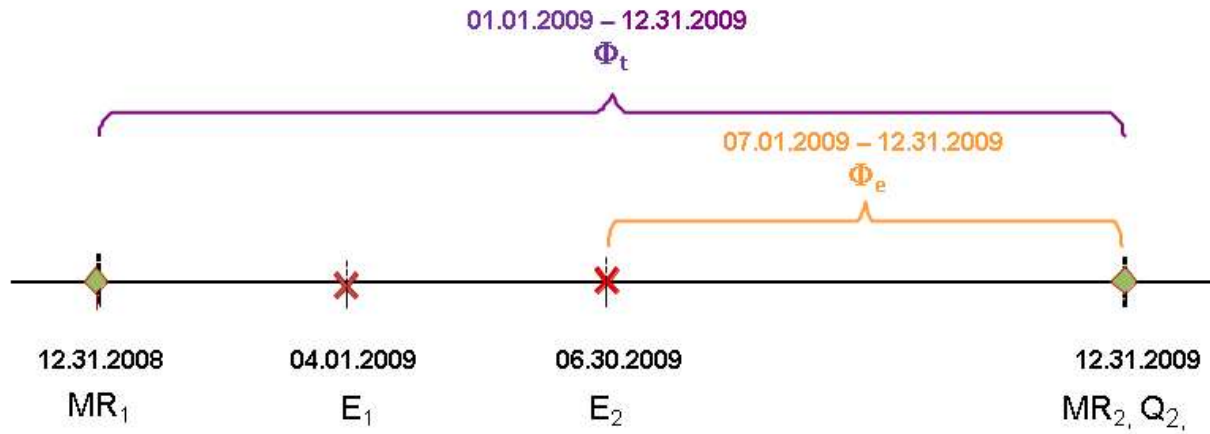
In this case, to harmonize the results, the gas factors are determined for the most recent actual previous meter reading result. The system calculates the quantity as follows:

$$Q_2 = (MR_2 - E_2) * \Phi_t$$

Formula

This is explained in the following figure.

- E = Estimated Meter Reading Result
- Q2 = Quantity for Meter Reading Result MR2
- Φ = Gas Factor



Example

In the example, E1 and E2 are estimated meter reading results. The gas factor Φ_e is an averaged value for the period E2-MR2. However, the gas factor Φ_e is not used for quantity determination for the period E2-MR2. The gas factor Φ_t , which relates to the period up to the next actual previous meter reading result, is used instead.

This procedure allows more consistency with regard to further actions in the period MR1-MR2, such as interpolation and deletion or correction of estimated meter reading results.